

Particle shape effects on dynamic properties of granular soils: A DEM study

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ABSTRACT

Particle shape is an essential aspect of granular soils that affects their mechanical properties. It is challenging to isolate the effect of particle shape from other mechanical properties such as inter-particle friction coefficient and particle shear modulus in laboratory tests. The discrete element method (DEM) makes it feasible to investigate the effect of macroscopic and mesoscopic particle morphologies including sphericity and roundness. This study provides a systematical DEM investigation of the particle shape effect on the dynamic properties of granular soils by simulating cyclic undrained triaxial tests at various initial confining pressures and void ratios. The results indicate that specimen consisting of angular particles generally exhibit a higher secant shear modulus G than that of rounded particles. Additionally, particle shape has no significant effect on modulus ratio G/G_0 but a noticeable impact on damping ratio D at relatively large strains due to dilatancy. A weighted coordination number CN' was proposed to reveal the microscopic effect of particle shape on G_0 . Contact sliding was found to be the main cause of modulus degradation and damping increase with increasing shear strain. The tangential shear modulus ratio G_{tan}/G_0 is inversely correlated to the weighted contact sliding ratio f_s' .

1. Introduction

Granular soils are the most common geomaterials and construction materials on earth. The dynamic properties of granular soils, including secant shear modulus G and damping ratio D , are essential in engineering analyses involving dynamic loads such as machine foundations design and earthquake ground response analysis. Generally, the shear modulus degrades and damping ratio increases with the increase of shear strain. At very small strains (e.g., below 1×10^{-6}), the shear modulus has the largest value termed as the small strain shear stiffness G_0 , while the damping ratio has the smallest value.

Numerous studies have been conducted to investigate factors affecting the dynamic properties of granular soils, which show that G_0 depends on void ratio (Hardin and Richart, 1963, Gu et al., 2013), mean effective stress (Hardin and Richart, 1963, Gu et al., 2013), soil gradation (Wichtmann and Triantafyllidis, 2009, Payan et al., 2017b), anisotropy (Wang and Mok, 2008, Gu et al., 2023), fines content (Payan et al., 2017a), and particle shape (Altuhafi et al., 2016, Payan et al., 2017b). Among these factors, the effect of particle shape is the most difficult to evaluate. Various shape indexes have been proposed to quantify the shape of soil particles. Sphericity (S) is proposed to describe

the overall angularity, and roundness (R) is proposed to capture the particle protrusions (Wadell, 1932). Aspect ratio (AR) is another commonly used shape index to capture the regularity of the particle which is defined as the ratio between Feret minimum and Feret maximum diameters (Altuhafi et al., 2013). Besides, convexity (C), which is defined as the ratio of the projected particle area to the gross area including all re-entrant sections, is proposed to capture the compactness of the particle (Altuhafi et al., 2016). On a smaller scale, roughness is proposed to capture surface texture which is quantified as the root mean square (RMS) deviation of the surface from its average level (RMS_f) (Schwarcz and Shane, 1969). A consensus among these studies was reached that the particle shape cannot be precisely described by a single shape index. Despite of the difficulty of accurately measuring the shape of a real particle, the most essential characteristics of particle shape can be captured by sphericity, roundness and roughness, which characterize the particle morphology on different scales (Mitchell and Soga, 2005, Blott and Pye, 2008, Fei et al., 2019, Tsigginos and Zeghal, 2019).

Moreover, many studies have tried to determine the effect of particle shape on G_0 using experimental techniques. Cho et al. (2006) and Sarkar et al. (2022) have carried out experiments using assemblies of similar

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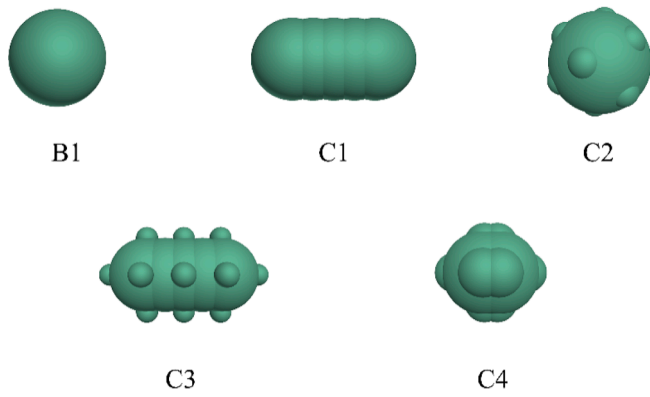


Fig. 1. Schematic of five selected particles.

Table 1
Particle label, type, composition, and shape indexes.

Particle label	Particle type	Composition	Sphericity (S)	Roundness (R)	Regularity (ρ)
B1	Sphere	Single sphere	1.00	1.00	1.00
C1	Clump	Five equal spheres	0.50	1.00	0.75
C2		A sphere (r_1) with 12 bumps ($r_2 = 0.286 r_1$)	0.92	0.29	0.61
C3		Five equal spheres (r_1) with 14 bumps ($r_2 = 0.3 r_1$)	0.43	0.30	0.37
C4		Two equal spheres (r_1) with 10 bumps ($r_2 = 0.5 r_1$)	0.74	0.5	0.62

relative densities. They found that G_0 increases with increasing sphericity or roundness of sand grains. Altuhafi et al. (2016) adopted void ratio, particle size and grading to normalize G_0 for better comparison. They found that the normalized G_0 increases when a specifically defined shape index termed as shape-angularity group indicator increases. However, Enomoto (2016) reported particle angularity has no

significant effect on modulus normalized by void ratio. Although the previous studies endeavored to experimentally investigate the effect of particle shape on small strain stiffness G_0 , definitive conclusion was not reached due to the difficulty in controlling particle shape as designed. Nevertheless, the discrete element method (DEM) provides a practical tool to study the effect of particle shape while maintaining other mechanical variables. Via DEM simulations, Tong and Wang (2014) and Tang and Yang (2021) found that G_0 increases with the increase of particle aspect ratio. Tsigginos and Zeghal (2019) reported that G_0 increases with increasing sphericity given a constant coordination number. Furthermore, Nie et al. (2022) showed that the small strain stiffness increases with increasing content of angular particles at the same void ratio and confining pressure.

In addition to small strain stiffness G_0 , another two essential dynamic properties of granular soils are modulus ratio G/G_0 and damping ratio D . The main factors affecting the modulus and damping ratio of granular soils are mean effective stress (Kokusho, 1980, Senetakis et al., 2012), shear strain (Kokusho, 1980), and particle gradation (Wichtmann and Triantafyllidis, 2013). However, previous studies have reached different conclusions about the effect of void ratio on modulus ratio G/G_0 and

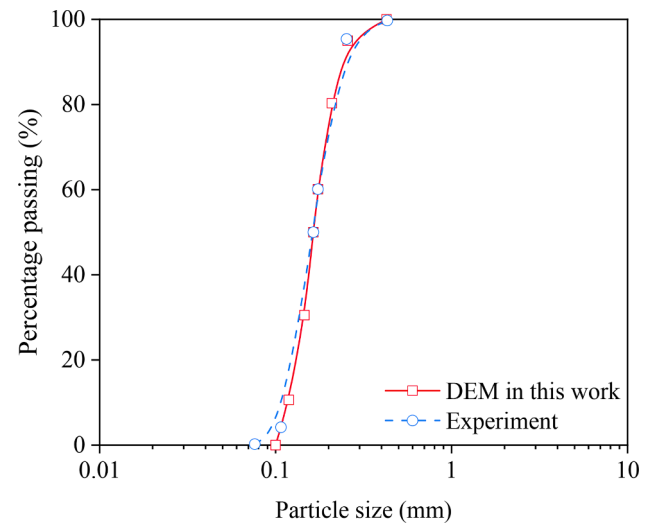


Fig. 3. Particle size distributions of Toyoura sand in experiments and simulations.

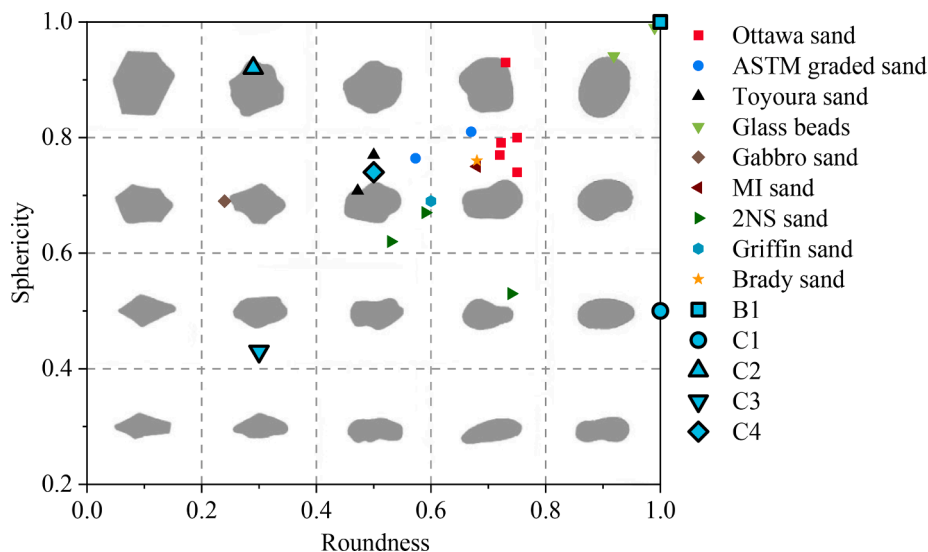


Fig. 2. Distribution of shape indexes of selected particles in the DEM simulations in comparison to typical sand particles (modified after Tsigginos and Zeghal, 2019).

Table 2
Simulation program in this study.

Particle number	Void ratio e	Relative density D_r	Confining pressure p_0 (kPa)	$e_{\max}/e_{\min}$	Shear strain
B1	0.589	99%	50	0.727/	1×10^{-6}
	0.588/	100%/	100	0.588	1×10^{-5}
	0.653/	53%/29%			1×10^{-4}
	0.687				1×10^{-3}
C1	0.587	100%	200		1×10^{-4}
	0.454	98%	50	0.698/	5×10^{-4}
	0.453/	99%/64%/	100	0.450	1×10^{-3}
	0.539/	42%			1×10^{-3}
C2	0.595	99%	200		
	0.453	97%	50	0.756/	
	0.555/	97%/55%/	100	0.549	
	0.642/	34%			
C3	0.685	98%	200		
	0.554	99%	50	0.910/	
	0.577/	99%/68%/	100	0.574	
	0.681/	46%			
C4	0.757	99%	200		
	0.576	99%	50	0.716/	
	0.466	100%/	100	0.464	
	0.633	58%/33%			
	0.464	100%	200		

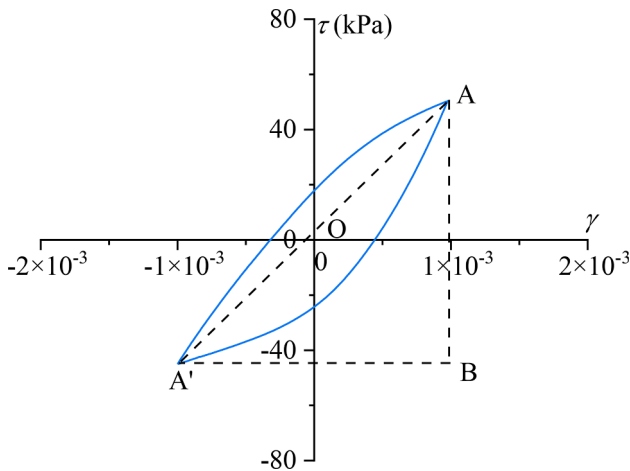


Fig. 4. Schematic of a typical hysteresis loop.

damping ratio D . Several experimental results indicated no significant effect of void ratio on G/G_0 and damping ratio D (Iwasaki et al., 1978, Tatsuoka et al., 1978, Kokusho, 1980, Senetakis et al., 2012, Senetakis et al., 2013, Wichtmann and Triantafyllidis, 2013), while Oztoprak and Bolton (2013) and Upreti and Leong (2021) found that different void ratios result in different G/G_0 and D at the same strain level and confining pressure. Although particle shape plays a crucial role in affecting small strain stiffness G_0 , its effect on modulus ratio G/G_0 and damping ratio D remains unclear. Chen et al. (2019) found that G/G_0 and D obtained from subrounded gravel-sand mixtures were significantly greater than and less than those of angular gravel-sand mixtures, respectively, at the same shear strain, relative density, and confining pressure. Nevertheless, the experiments conducted by Gu (2012) show no significant difference of modulus ratios G/G_0 for spherical glass beads and completely decomposed angular granite. Further comprehensive investigations are desired to fully understand the effect of particle shape on the modulus ratio G/G_0 and damping ratio D .

Numerous studies have explored the relationship between dynamic

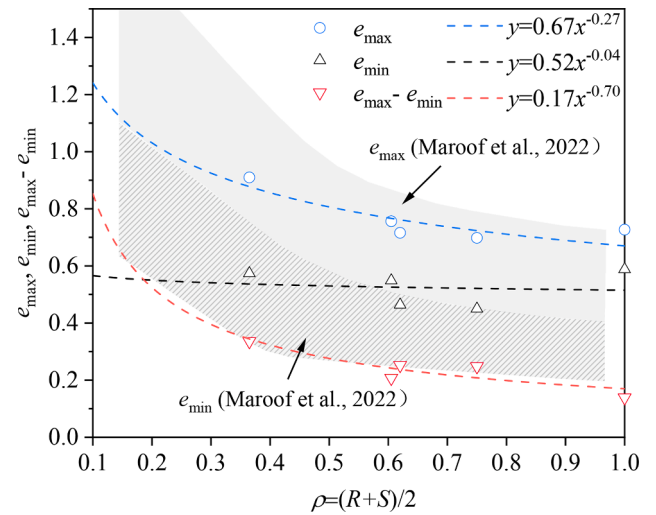


Fig. 5. Variation of $e_{\max}$, $e_{\min}$ and $e_{\max} - e_{\min}$ with regularity ρ .

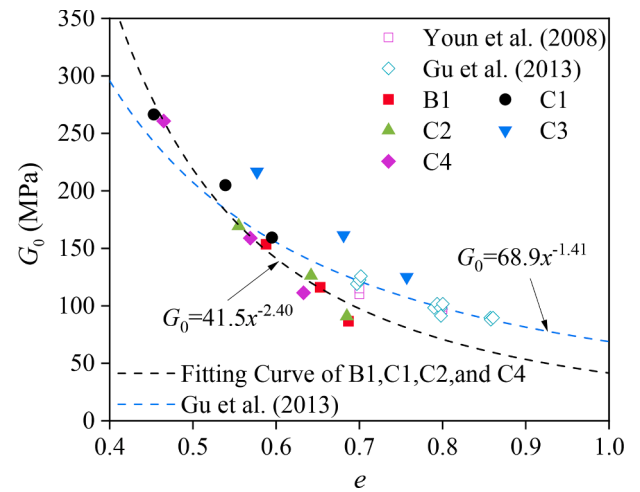


Fig. 6. Variation of G_0 with e for five different particle shapes at p_0 of 100 kPa in DEM simulations and experiments.

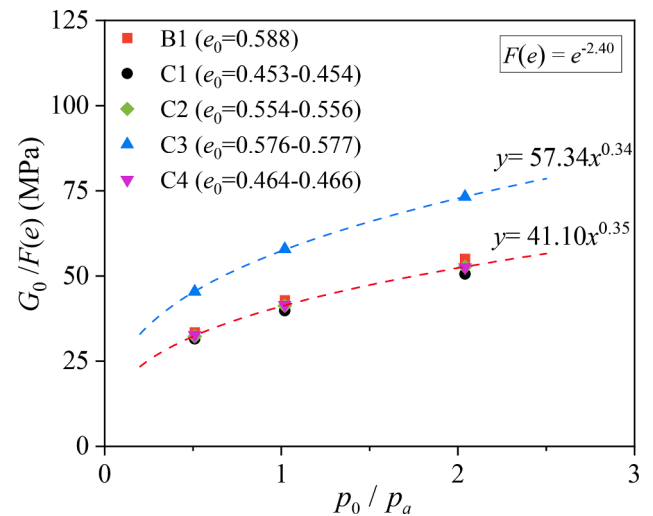


Fig. 7. Variation of $G_0/F(e)$ with p_0/p_a for five different particle shapes.

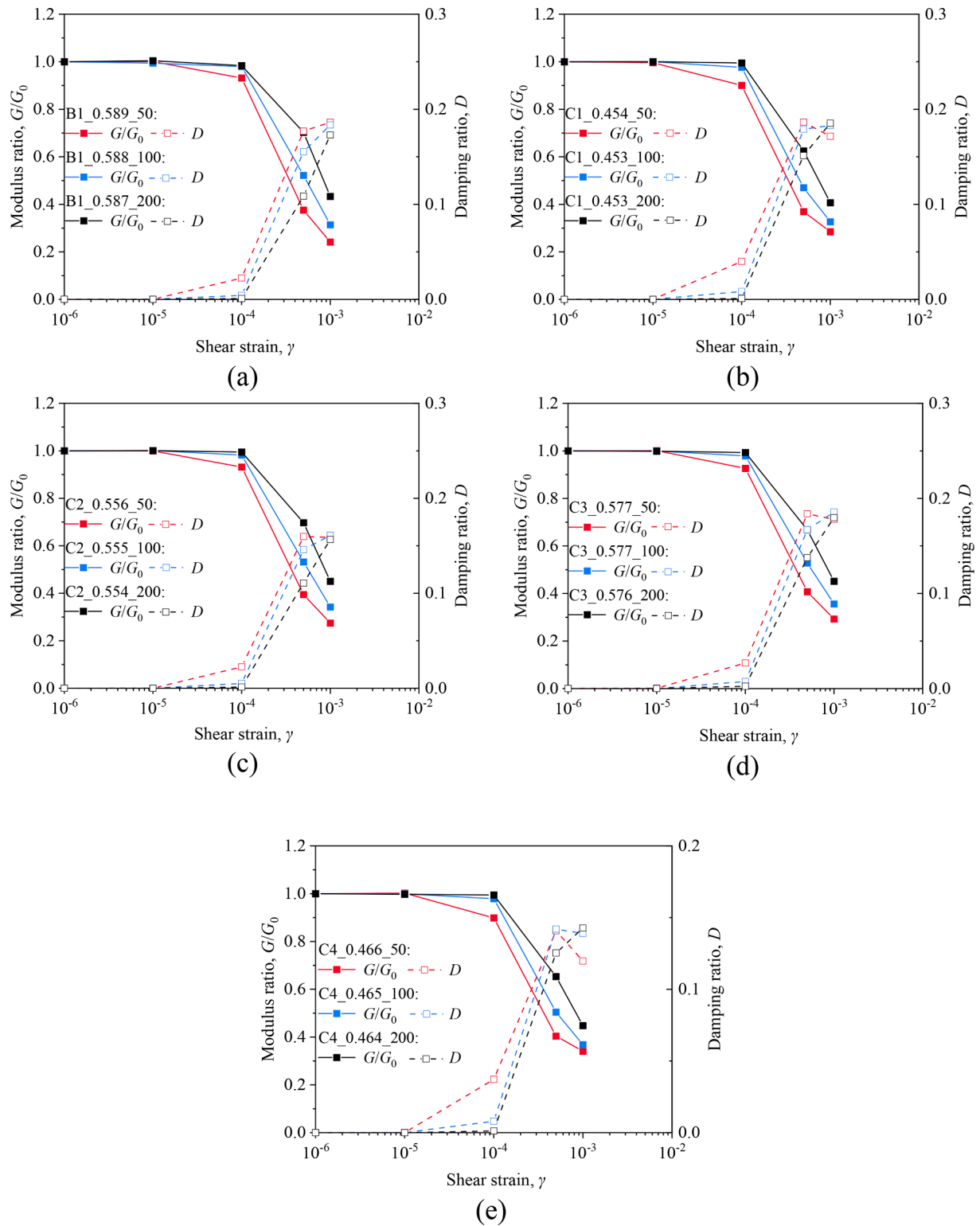


Fig. 8. Variation of G/G_0 and D with shear strain γ at similar initial void ratios under different confining pressures for different particle shapes: (a) B1; (b) C1; (c) C2; (d) C3; (e) C4.

properties and microscopic properties using DEM. [Gu and Yang \(2013\)](#), [Otsubo et al. \(2022\)](#) and [Liu et al. \(2022\)](#) discovered a strong correlation between the small-strain shear modulus and the mechanical coordination number for spherical particle assemblies. [Nie et al. \(2020\)](#) investigated the normalized small strain shear modulus $G_0/(P_0)^{1/3}$ and the dimensionless number $(Z_m/e)^{2/3}$ for five granular samples with varying particle shape characteristics. [Reddy et al. \(2022\)](#) examined the

variation of coordination number, fabric anisotropy, and contact force with increasing shear strain. However, these studies did not identify the primary cause for the degradation of modulus. Motivated by the review above, this study presents a DEM study to investigate the effect of particle shape on the dynamic properties of granular soils. Five different particle shapes with varying sphericities and roundness were considered for DEM simulations. Cyclic undrained triaxial tests were simulated to

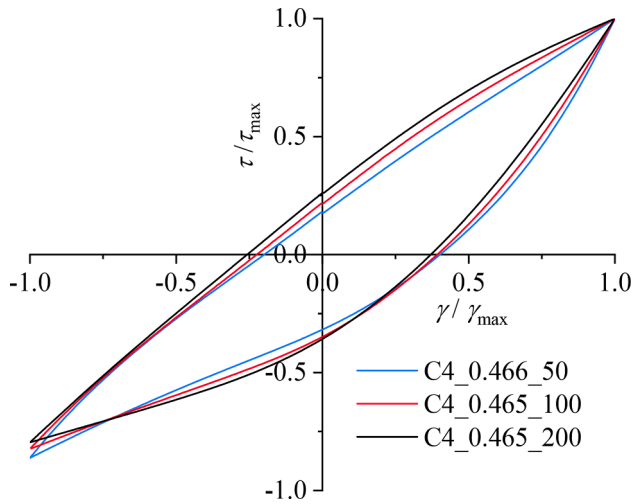


Fig. 9. Hysteresis loops of the assembly of particle C4 with similar initial void ratios under p_0 of 50, 100 and 200 kPa at $\gamma = 1 \times 10^{-3}$.

obtain the secant shear modulus G and damping ratio D . The effects of confining pressure, void ratio, and particle shape were investigated with both macroscopic and microscopic analyses.

2. DEM simulation

2.1. Particle shape indexes

This study focuses on the effect of particle shape on the dynamic properties of granular soils. Five distinct particles, with varying degrees of sphericity and roundness, were selected for analysis and are illustrated in Fig. 1. The particle label, type, composition, and shape indexes are listed in Table 1. The effect of surface texture (or roughness) was not considered in this work for simplicity since it was specially investigated in the previous experimental and numerical studies (Otsubo et al., 2015, Otsubo and O'Sullivan, 2018, Sandeep et al., 2021, Reddy et al., 2022). In this study, the particle shape only refers to the mesoscopic and macroscopic morphologies of particles.

The sphericity S and roundness R of a given particle was determined as follows (Tsigginos and Zeghal, 2019):

$$S = \frac{r_{\max} - r_{\min}}{r_{\max} - r_{\min} - r_{\text{cir}}} \quad (1)$$

$$R = \frac{\sum r_i / N}{r_{\max} - r_{\min}} \quad (2)$$

where $r_{\max}$ and $r_{\min}$ are the diameters of the largest inscribed and smallest circumscribed spheres and $\sum r_i / N$ is the average radius of the sharpness of angular protrusions from the particle.

The regularity ρ is defined as (Cho et al., 2006):

$$\rho = \frac{R + S}{2} \quad (3)$$

Eqs. (1) and (2) indicate that the sphericity is mainly determined by the particle elongation while the roundness is primarily determined by the particle sharpness of angular protrusions (Mitchell and Soga, 2005). Fig. 2 shows the distribution of shape indexes of five selected particles compared with the representative real sand particles. The sphericity of real sands ranges from 0.5 to 1.0 while the value of roundness varies from 0.2 to 0.8, which agrees with the result in Vepraskas and Cassel (1987). The ranges of sphericity and roundness of the selected particles encompass the major distribution of shape indexes of the interested sand particles.

2.2. Contact model and model parameters

In this work, Hertz-Mindlin contact model was adopted in the simulations. The particle shear modulus G_g was set to 18 GPa according to the experimental data for quartz sand particles (Sandeep et al., 2018). The particle Poisson's ratio ν_g and inter-particle friction coefficient μ were selected to be 0.15 and 0.5, respectively, according to Gu et al. (2020a).

2.3. Specimen generation

The specimen was an assembly of around 8000 particles in a cube contained by six rigid walls. The particle dimension of a non-spherical particle is measure by its volume equivalent spherical diameter, which is defined as the diameter of a sphere of equivalent volume. Fig. 3 shows the particle size distributions of Toyoura sand and the numerical sample. The ratios of the specimen size to the maximum particle equivalent spherical diameter for all the particle shapes are above 17.5 to account for the sample size effect, which is larger than the usually adopted ratios for non-spherical particle shapes (McDowell and Li, 2016, Gong and Liu, 2017, Nie et al., 2021, Nie et al., 2022).

The specimen was generated as follows. Firstly, particles were generated with the inter-particle friction coefficient μ set to zero. Then, the specimen was subjected to a confining stress of 10 kPa and cycled to reach equilibrium, with three inter-particle friction coefficients of 0.001, 0.1, and 0.2 used to generate specimens of different void ratios. The inter-particle friction coefficient μ was subsequently set to 0.5 in further consolidation and shearing. The gravity is neglected in this study.

2.4. Cyclic triaxial tests

Undrained cyclic triaxial tests were simulated in this study according to ASTM (2011). In undrained simulations, each wall moved at the designated velocities ensuring a constant volume (Shafipour and Soroush, 2008, Huang et al., 2018, Barnett et al., 2020, Keishing and Hanley, 2020). The strain-controlled method was adopted and the loading direction was alternated once the target shear strain was reached. As suggested by Huang et al. (2018) and Ishihara (1996), the loading frequency was set to 5 Hz. The inertial number I was smaller than 0.001 throughout the tests for maintaining a quasi-static state (da Cruz et al., 2005, Lopera Perez et al., 2016). Both interparticle and local damping coefficients were set to zero during cyclic shearing. Moreover, the timestep was increased by ten times to expedite the simulations by setting the mass density of the particles as $2.64 \times 10^5 \text{ kg/m}^3$ (Thornton, 2000, Tsigginos and Zeghal, 2019). A total number of 125 simulation tests were conducted for five different particle shapes at different strain levels, void ratios, and confining pressures. The first complete hysteresis loop was analyzed to obtain the secant shear modulus G and the damping ratio D . Table 2 presents the complete simulation program in this study. The maximum void ratio was obtained at a confining pressure of 10 kPa using a inter-particle friction coefficient of 0.5, while the minimum void ratio was obtained at the same pressure using a inter-particle friction coefficient of zero (Gu et al., 2020b, Wu et al., 2022). Note that the inter-particle friction of 0.5 was adopted in all the post-consolidation cyclic shearing tests for comparison purpose.

For DEM simulations, the shear stress τ and shear strain γ are defined as

$$\tau = \frac{\sigma'_v - \sigma'_h}{2} \quad (4)$$

$$\gamma = \varepsilon_v - \varepsilon_h \quad (5)$$

where σ'_v and σ'_h are the effective normal stresses in the vertical and horizontal directions respectively, and ε_v and ε_h are the vertical and horizontal strains respectively.

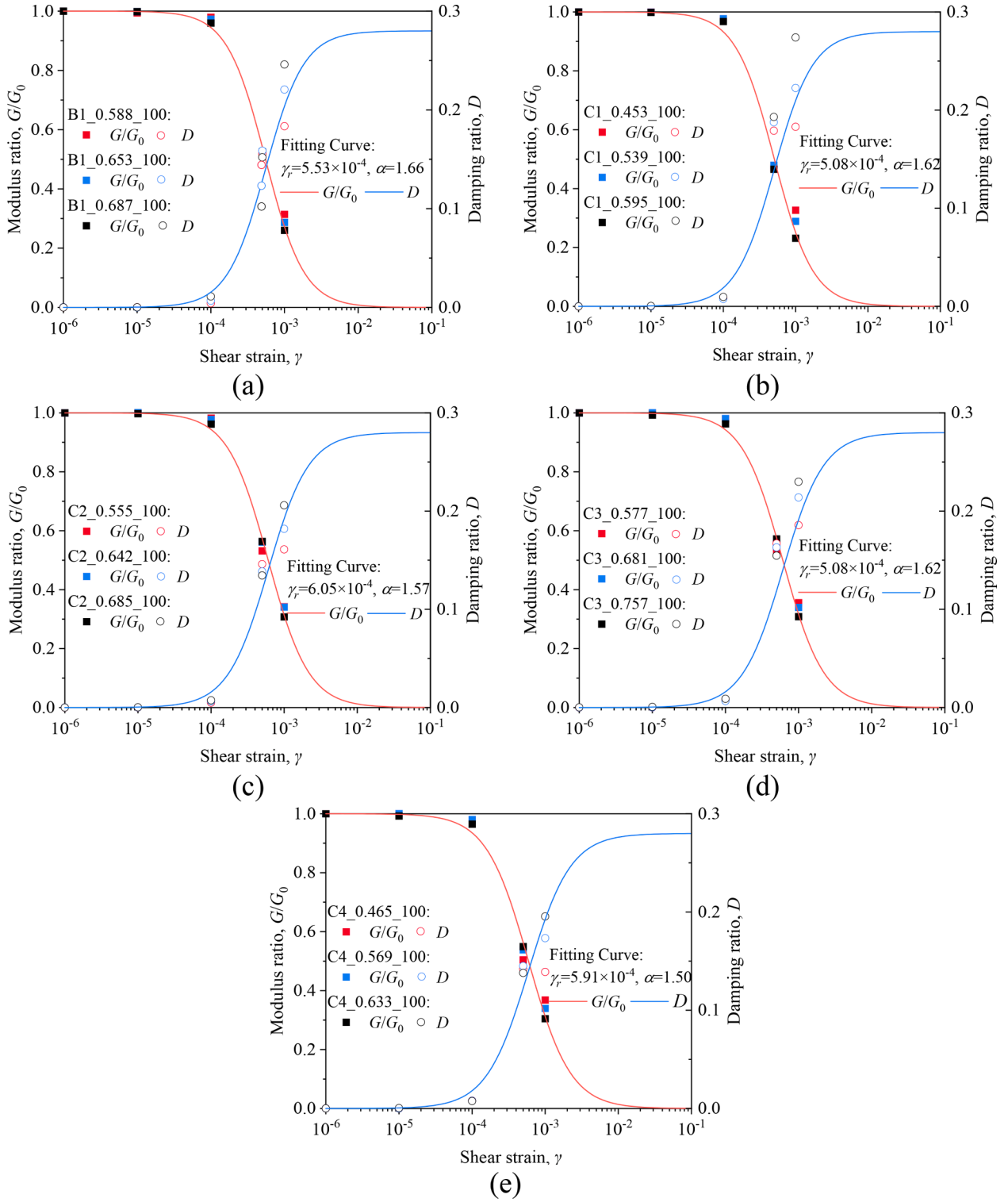


Fig. 10. Variation of G/G_0 and D with shear strain γ at p_0 of 100 kPa for different void ratios using five different particle shapes: (a) B1; (b) C1; (c) C2; (d) C3 and (e) C4.

Fig. 4 shows a schematic of a typical hysteresis loop. The shear modulus and damping ratio can be determined from the loop with the following equations (Kokusho, 1980)

$$G = \frac{y_A - y_{A'}}{x_A - x_{A'}} \quad (6)$$

$$D = \frac{1}{\pi} \frac{S_{\text{loop}}}{S_{\triangle AA'B}} \quad (7)$$

where y_A ($y_{A'}$) and x_A ($x_{A'}$) are the horizontal and vertical coordinates of the point A (A'), S_{loop} is the area of the whole hysteresis loop, and $S_{\triangle AA'B}$ is the area of the triangle $AA'B$.

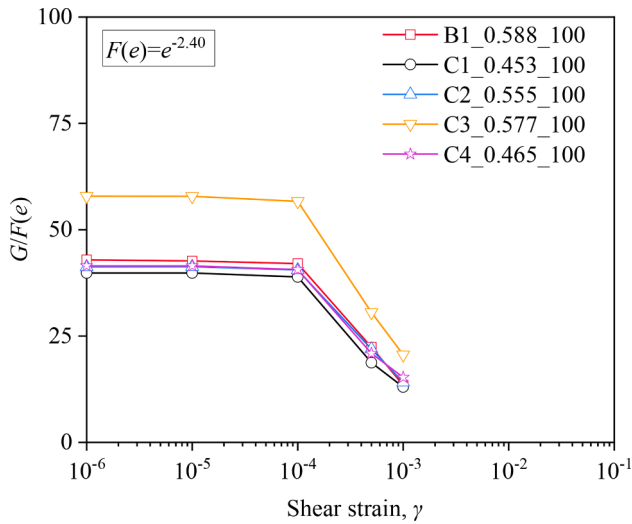


Fig. 11. Variation of $G/F(e)$ with shear strain γ at p_0 of 100 kPa using five different particle shapes.

3. Results and discussions

3.1. Macroscopic properties

3.1.1. Maximum void ratio $e_{\max}$ and minimum void ratio $e_{\min}$

Fig. 5 shows the variation of $e_{\max}$, $e_{\min}$, and $e_{\max}-e_{\min}$ with regularity ρ based on DEM simulations, together with the experimental data from Maroof et al. (2022). The result shows that the maximum void ratio decreases and the minimum void ratio slightly decreases with the increase of regularity. The variation of the maximum void ratio $e_{\max}$ is similar to that in Maroof et al. (2022), but the range of the minimum void ratio $e_{\min}$ is much narrower than experimental data. Previous experimental work reported that sphere assemblies have smaller maximum and minimum void ratios than angular particles (Altuhafi et al., 2016), while the sphere assembly, with the highest regularity, can exhibit slightly higher maximum and minimum void ratios than angular particles, as reported in the numerical study by Garcia and Bray (2019). This discrepancy exists because spherical particles (e.g., glass beads) tend to have lower inter-particle friction coefficients, which leads to a lower void ratio, but the friction coefficients are the same for particles with different shapes in the DEM simulations. Besides, it can be observed that the void ratio range $e_{\max}-e_{\min}$ decreases with increasing ρ , which is consistent with the experimental data (Cho et al., 2006, Maroof et al., 2022).

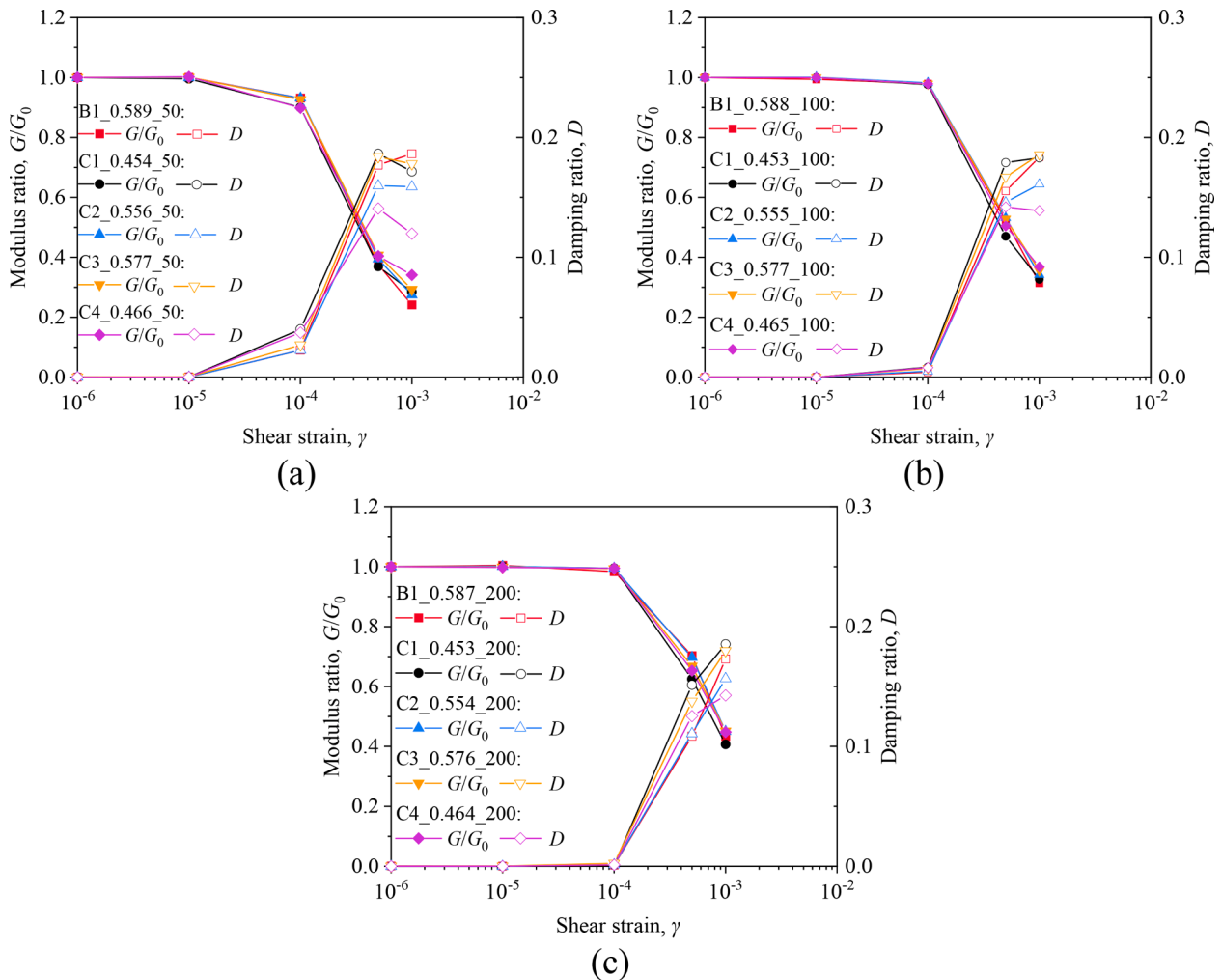


Fig. 12. Variation of G/G_0 and D with shear strain γ using five different particle shapes for dense assemblies at (a) 50 kPa; (b) 100 kPa and (c) 200 kPa.

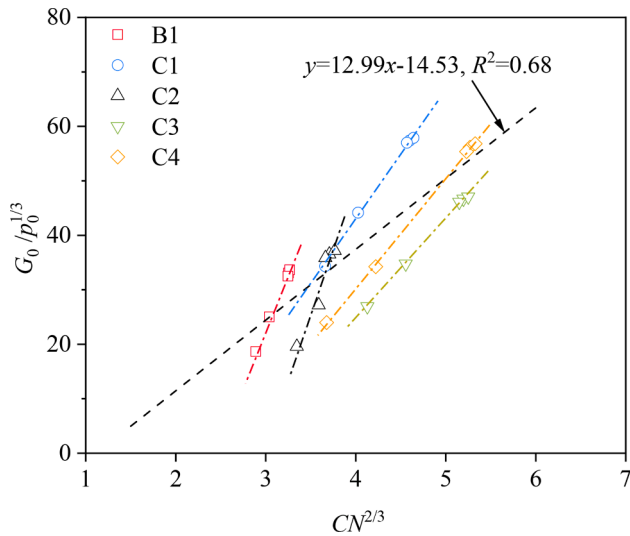


Fig. 13. Correlation between stress normalized shear modulus (G_0 in MPa, p_0 in kPa) and coordination number (CN) for five different particle shapes.

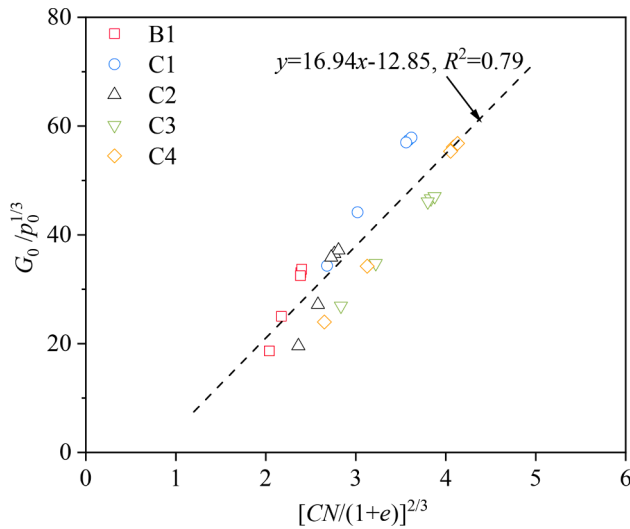


Fig. 14. Correlation between the stress normalized shear modulus (G_0 in MPa, p_0 in kPa) and $[CN/(1+e)]^{2/3}$ for five different particle shapes.

3.1.2. Small strain stiffness G_0

It is well accepted that the G_0 can be expressed in a power function form as (Hardin and Richart, 1963)

$$G_0 = AF(e) \left(\frac{p_0}{p_a} \right)^n \quad (8)$$

where A is a constant reflecting soil type, grain properties and fabric, p_0 is the confining pressure, p_a is a reference stress (98 kPa in this study), n is a stress exponent reflecting the effect of confining pressure and $F(e)$ is a void ratio function reflecting the effect of soil density. Lo Presti et al. (1997) proposed a widely used form of $F(e)$ as

$$F(e) = e^{-x} \quad (9)$$

Fig. 6 shows the variation of G_0 with e at p_0 of 100 kPa in DEM simulations, together with the experimental data for Toyoura sand in Youn et al. (2008) and Gu et al. (2013). The value of G_0 decreases as the void ratio increases, as expected. The results for assemblies comprising particles B1, C1, C2, or C4 can be fitted by the same curve while the assembly of particle C3 exhibits a much larger value of G_0 . It indicates

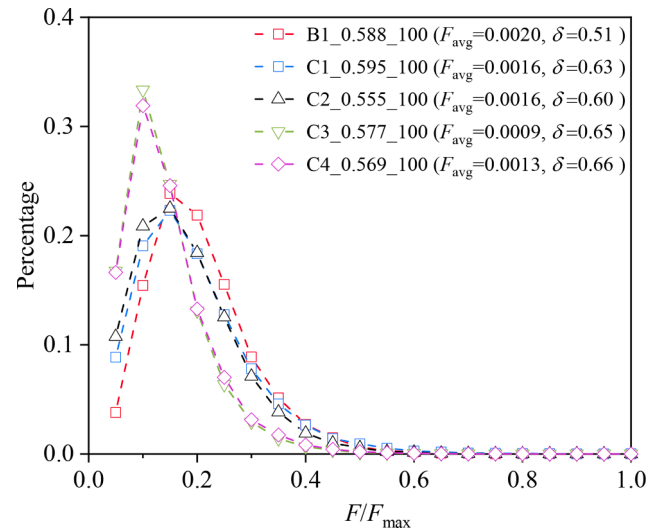


Fig. 15. Distribution of the normalized contact normal force $F/F_{\max}$ for five particle shapes at p_0 of 100 kPa with e from 0.555 to 0.595.

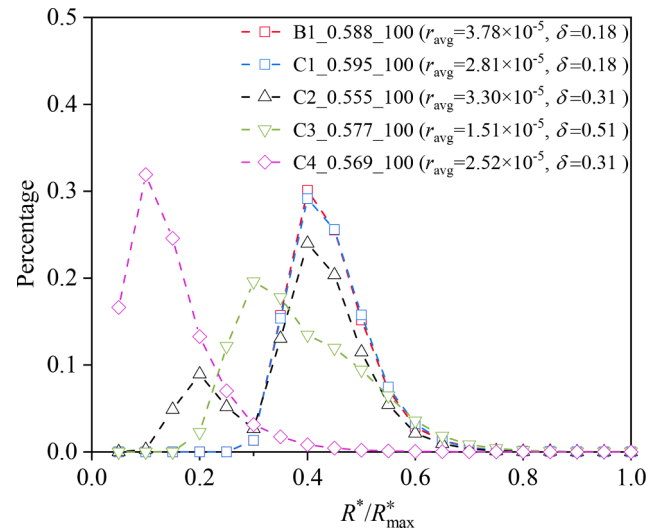


Fig. 16. Distribution of the normalized contact radius $R^*/R_{\max}^*$ for five particle shapes at p_0 of 100 kPa with e from 0.555 to 0.595.

the difference observed among the B1, C1, C2, and C4 assemblies is only caused by the void ratio, whereas the distinctions between the C3 assembly and the other four is owing to the drastically different regularities. The void ratio function fitted with the experiment data is also added to the figure for comparison (Gu et al., 2013).

Fig. 7 plots the G_0 values normalized by the void ratio function $F(e)$ against the normalized pressure p_0/p_a for five different particle shapes. Normalized shear modulus increases with the increase of confining pressure as expected. In this study, the fitted value for the exponent x in the void ratio function $F(e)$ is 2.40. The result shows that the shear moduli of specimens with different particle shapes except for particle C3 can be normalized using the same void ratio function (Eq. (9)), while the specimen of particle C3 exhibits a much larger normalized shear modulus due to its low regularity. Meanwhile, the stress exponents n are almost the same (i.e., ranging from 0.34 to 0.35) for specimens with different particle shapes. It suggests that the particle shape does not affect the stress dependence of G_0 in this study.

3.1.3. Modulus ratio G/G_0 and damping ratio D

Fig. 8 shows the effect of confining pressure on G/G_0 and D at similar

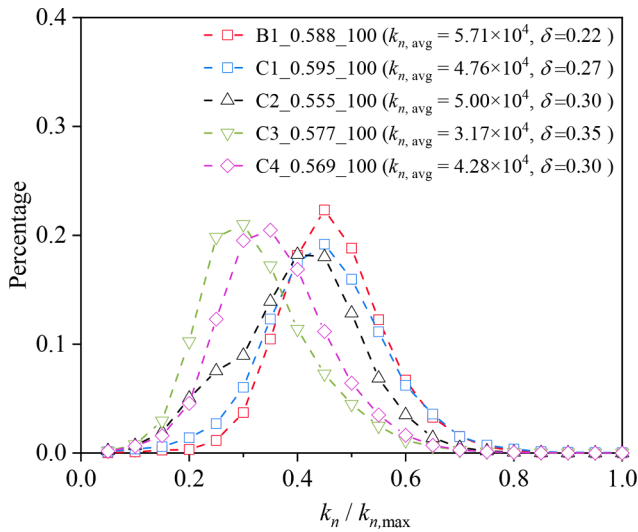


Fig. 17. Distribution of the normalized contact normal stiffness $k_n/k_{n,max}$ for five particle shapes at p_0 of 100 kPa with e from 0.555 to 0.595.

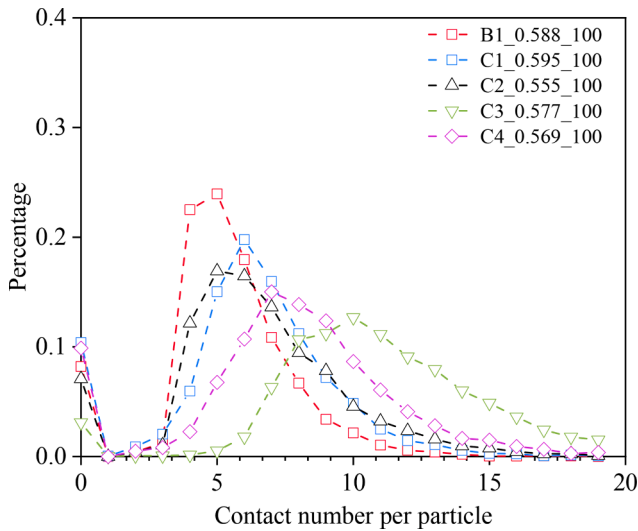


Fig. 18. Distribution of the contact number per particle for five different particle shapes at p_0 of 100 kPa with e from 0.555 to 0.595.

void ratios for different particle shapes. Note that the specimen labels sequentially refer to particle type, initial void ratio, and initial confining pressure. For example, B1_0.589_50 is a specimen comprising particle B1 with e_0 of 0.589 under an initial confining pressure of 50 kPa. The results show that the modulus ratio G/G_0 increases as the confining pressure p_0 increases at the same strain level. Note that the inter-particle friction coefficient is an important factor ruling the stiffness degradation, i.e. a lower friction coefficient is supposed to result in faster degradation of modulus ratio G/G_0 with the increasing shear strain (Reddy et al., 2022), while the microscopic morphology is not considered in this study for simplicity and the friction coefficient of 0.5 was adopted for all the simulations. The damping ratio D increases with the decrease of confining pressure p_0 at strain levels less than 1×10^{-3} . However, the damping ratios of particle C1, C3, and C4 do not keep increasing with the decrease of confining pressure p_0 after the strain level reaches 1×10^{-3} , since the specimen under a relatively low confining pressure is more dilatant at a large strain and therefore is of a lower damping ratio. Fig. 9 shows the hysteresis loops of the assembly of particle C4 with similar initial void ratios under p_0 of 50, 100 and 200 kPa at $\gamma = 1 \times 10^{-3}$. The normalized stress τ/τ_{max} and normalized strain γ/γ_{max} were

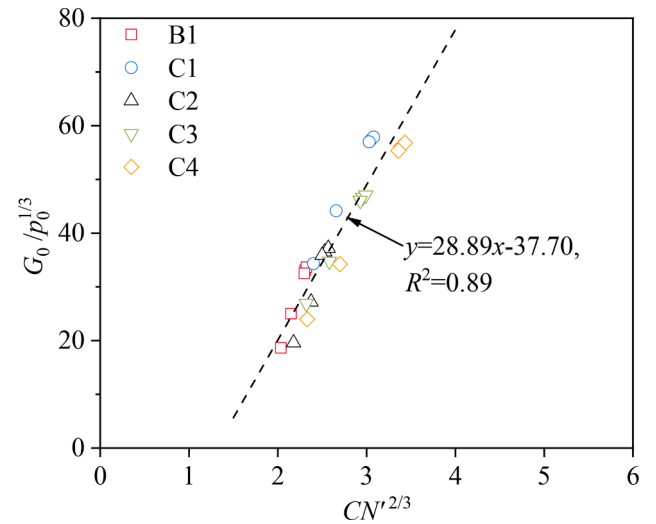


Fig. 19. Correlation between the stress normalized shear modulus (G_0 in MPa, p_0 in kPa) and modified coordination number CN' for five different particle shapes.

used for better comparison. The result indicates that the hysteresis loops become more asymmetrical as the confining pressure decreases due to dilation. Furthermore, the loop area tends to decrease with the decreasing confining pressure, which indicates a decreasing damping ratio D .

The commonly adopted hyperbolic modulus degradation and damping models proposed by Darendeli (2001) and Hardin and Drnevich (1972) are

$$\frac{G}{G_0} = \frac{1}{1 + (\gamma/\gamma_r)^\alpha} \quad (10)$$

$$\frac{D}{D_0} = 1 - \frac{G}{G_0} \quad (11)$$

where γ_r is the reference shear strain, α is the curvature parameter, G_0 is the maximum shear modulus and D_0 is the maximum damping ratio. The maximum damping ratio is assumed to be 0.28 in this work according to Hardin and Drnevich (1972).

Fig. 10 shows the effect of void ratio on modulus ratio G/G_0 and damping ratio D at p_0 of 100 kPa for five different particle shapes. It indicates that the effect of void ratio on modulus ratio G/G_0 is insignificant, which is consistent with the experimental observations (Iwasaki et al., 1978, Tatsuoka et al., 1978, Kokusho, 1980, Senetakis et al., 2013, Wichtmann and Triantafyllidis, 2013). Besides, no significant effect of void ratio is found on D at a strain level less than 1×10^{-3} . However, loose specimen tends to have a higher damping ratio than dense specimen after the strain level exceeds 1×10^{-3} . Dense specimen is dilatant which results in a smaller D as explained before while the loose specimen is contractive which leads to the decrease of effective confining pressure during shearing and therefore the increase of damping ratio. Furthermore, the effects of dilatancy and contraction have more significant effects on damping ratio D than modulus ratio G/G_0 . Additionally, the fitting results of modulus and damping ratios are also shown in the figure using Eqs. (10) and (11). Eq. (10) shows a good fitting performance while the fitting result of Eq. (11) is not satisfactorily good in comparison especially at $\gamma = 1 \times 10^{-3}$ because the dilatancy and contraction of the specimens are not well accounted in these two equations.

Fig. 11 shows the effect of particle shape on $G/F(e)$ at p_0 of 100 kPa but different shear strains γ . The assembly comprising particle C3, which has the lowest sphericity and roundness, shows a remarkably larger normalized secant shear modulus $G/F(e)$ than other assemblies, while

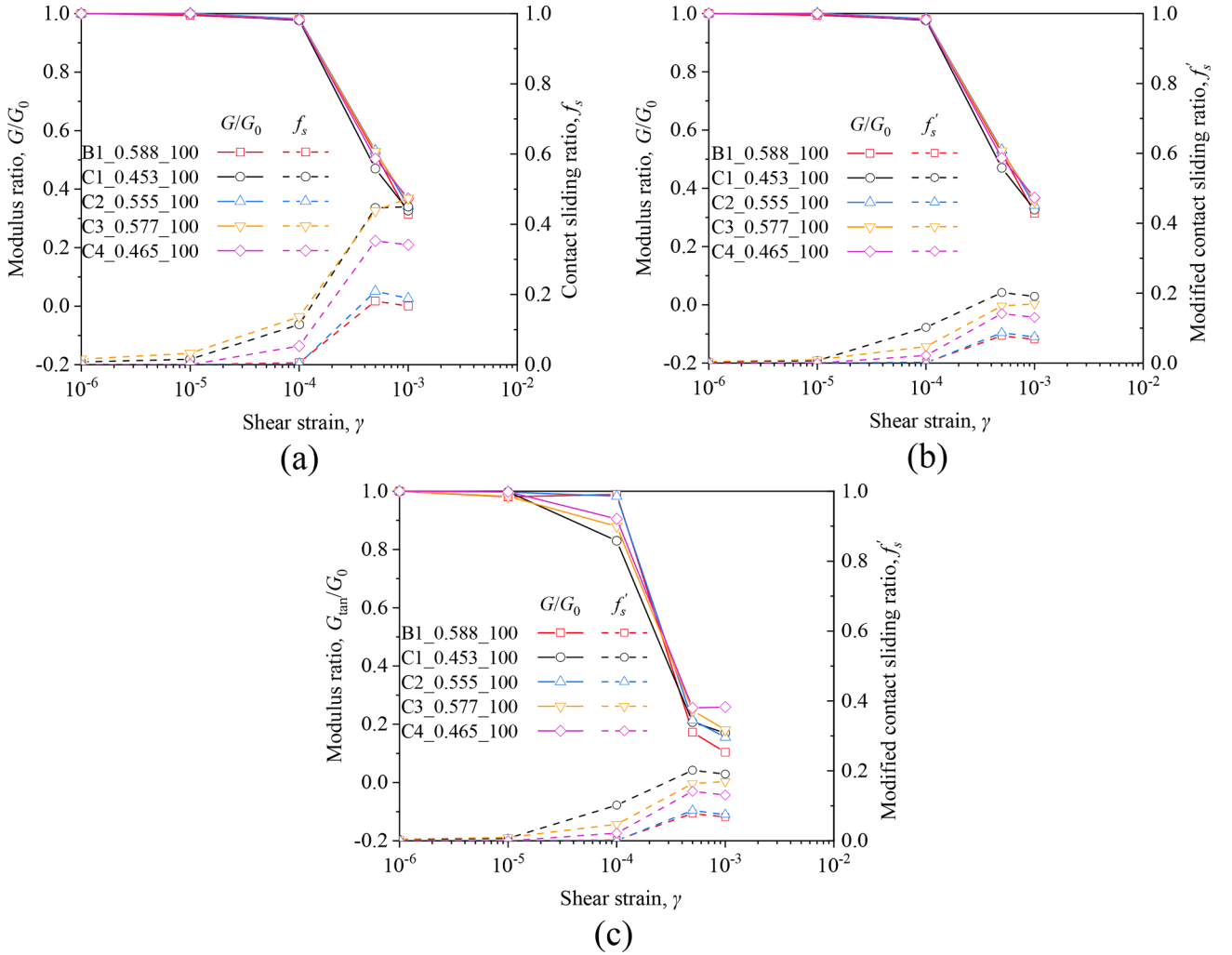


Fig. 20. A comparison between: (a) secant modulus ratio G/G_0 and contact sliding ratio f_s , (b) secant modulus ratio G/G_0 and modified contact sliding ratio f'_s , and (c) tangential modulus ratio G_{tan}/G_0 and modified contact sliding ratio f'_s .

those of B1, C1, C2, and C4 exhibit similar $G/F(e)$. This result suggests that the particle shape has a remarkable effect on secant shear modulus G which is consistent with the experimental data (Cho et al., 2006) and this effect is not only determined by the void ratios of assemblies with different particle shapes. Fig. 12 shows the effects of particle shape on G/G_0 and D at p_0 of 50, 100, and 200 kPa for dense assemblies. The result indicates that the modulus ratio G/G_0 is almost independent of particle shape, but the particle shape does affect damping ratio D significantly because sample consisting of irregular particles has stronger dilatancy than that of rounded particles (Xiao et al., 2019). The result shows that dilatancy of the specimens increases with the decrease of regularity. The damping ratio is similar when the strain level is less than 1×10^{-4} since dilatancy does not have a significant effect at a small strain. However, when the shear strain reaches 5×10^{-4} , dilatancy exerts its influence on damping ratio which begins to decrease with the increase of shear strain at 50 and 100 kPa.

3.2. Micromechanical interpretations

The coordination number is a widely used microscopic index of granular materials. As suggested by Gu and Yang (2013), the stress normalized secant shear modulus $G_0/(p_0)^{1/3}$ is plotted against $CN^{2/3}$ for five different particle shapes in Fig. 13, where coordination number CN is defined as

$$CN = \frac{2(C - N_1)}{N - N_1 - N_0} \quad (12)$$

where C is the number of contacts, N is the number of particles, N_1 is the number of particles with only one contact, and N_0 is the number of particles without any contact.

It shows that $G_0/(p_0)^{1/3}$ is linear related to $CN^{2/3}$ for all the assemblies with all the five considered particle shapes. It can be seen that the slopes of these lines are different. The relationship for assemblies comprising particles with lower sphericity (i.e., C1, C3 and C4) tends to have a flatter slope. The coefficient of determination R^2 is 0.68, which is worse than the result for spheres (Gu and Yang, 2013, Gu et al., 2017, Gu and Yang, 2018). Meanwhile, assemblies of angular particles with higher coordination numbers have even smaller stress normalized shear moduli than the assembly of spheres. To construct a better correlation between the microscopic and macroscopic properties, additional parameters are needed to eliminate the effect of particle shapes.

According to Chang et al. (1989), the small strain shear modulus G_0 can be determined by the void ratio and several microscopic properties by Eq. (13).

$$G_0 = \frac{1}{5} \left[\frac{\sqrt{3}}{\sqrt{2\pi(1-\nu_g)}} \right]^{\frac{2}{3}} \left(\frac{5-4\nu_g}{2-\nu_g} \right) \left[\frac{CN}{1+e} \right]^{\frac{2}{3}} G_g^{\frac{1}{3}} p_0^{\frac{1}{3}} \quad (13)$$

where G_g is the particle shear modulus, ν_g is the particle Poisson's ratio,

CN is the coordination number of the assembly, e is the void ratio of the assembly and p_0 is the confining pressure. Motivated by the above equation, the dimensionless number $CN/(1+e)$ is introduced to establish a unique correlation between microscopic and macroscopic characteristics. Fig. 14 presents the correlation between the stress normalized modulus $G_0/(p_0)^{1/3}$ and $[CN/(1+e)]^{2/3}$ for five different particle shapes. A better relationship can be observed suggesting that Eq. (13) is also applicable to angular particle assemblies. The coefficient of determination R^2 is 0.79 which is better than the result in Fig. 13.

The distribution of contact associated information is an essential factor affecting G_0 besides magnitudes of contact stiffness and coordination number (Gu and Yang, 2013). For Hertzian contact, the contact normal stiffness k_n (Johnson, 1987) is

$$k_n = \left[\frac{2G_g}{3(1-\nu_g)} \right]^{2/3} (R^*)^{1/3} F_n^{1/3} \quad (14)$$

where R^* is defined as $1/R^* = 1/R_1 + 1/R_2$ with R_1 and R_2 the radii of two contacting particles.

Fig. 15 presents the distribution of the normalized contact normal force $F/F_{\max}$ for five particle shapes at p_0 of 100 kPa with e from 0.555 to 0.595. The standard deviation δ for contact associated information can be obtained using the equation given below.

$$\delta = \sqrt{\frac{\sum_{i=1}^N (X_i/X_{\text{avg}} - 1)^2}{N}} \quad (15)$$

where X_i is the magnitude of a specific property quantity at the i th contact and N is the total number of contacts, X_{avg} is the average magnitude of the property. Fig. 15 shows that as sphericity or roundness decreases, the curve shifts to the left with the average contact force decreasing and the standard deviation δ of normalized contact normal force $F/F_{\max}$ increasing slightly. Fig. 16 shows the distribution of the normalized contact radius $R^*/R_{\max}^*$ for five particle shapes at p_0 of 100 kPa with e from 0.555 to 0.595. The result indicates that lower sphericity and roundness both contribute to a higher standard deviation of normalized contact radius $R^*/R_{\max}^*$ and a smaller average contact radius. Note that two peaks are observed for the assembly of particle C2 since it has two dominant radii which are very distinct from each other.

Fig. 17 shows the distribution of the normalized contact normal stiffness $k_n/k_{n,\max}$ for five particle shapes at p_0 of 100 kPa with e from 0.555 to 0.595. As regularity decreases, the curve shifts to the left with the average normal contact stiffness decreasing and the standard deviation of normalized contact normal stiffness $k_n/k_{n,\max}$ increasing. This result is consistent with Tang and Yang (2021). Given the results shown in Figs. 15 and 16, it can be concluded that the distribution of normal contact stiffness mainly corresponds to the distribution of contact radius rather than that of contact force.

Fig. 18 shows the distribution of the contact number per particle for five different particle shapes at p_0 of 100 kPa with e from 0.555 to 0.595. It indicates that the number of floating particles, i.e., particles with no contacts, in the assembly comprising particle C3 is smaller than that in other assemblies. Although the floating particles do not contribute to the instantaneous bearing capacity of the assembly at small strain level, they contribute to the void ratio, which explains the reason that the specimen of particle C3 has larger maximum and minimum void ratios and also larger normalized small strain shear modulus as shown in Fig. 11. Moreover, the percentage of particles with small amount of contacts is relatively small which is consistent with the results in Gu et al. (2017). For the particles with more than four contacts, the curves shift to the right as particle regularity decreases. The result suggests that the distribution of particle-wise coordination number becomes less uniform as particle regularity decreases.

To account for the non-uniform distribution of normal contact stiffness, a weighted factor $[(F_i/F_{\text{avg}})(R_i^*/d_{50})]^{1/3}$ is introduced in the

calculation of a weighted coordination number, which is denoted as modified coordination number CN' and defined as

$$CN' = 2 \frac{\sum_{i=1}^C [(F_i/F_{\text{avg}})(R_i^*/d_{50})]^{1/3} - \sum_{i=1}^{C_1} [(F_i/F_{\text{avg}})(R_i^*/d_{50})]^{1/3}}{N - N_1 - N_0} \quad (16)$$

where C_1 is the number of contacts for which at least one of the contacting particles has one contact, F_i is the normal force of the i th contact, R_i^* is the contact radius of the i th contact, and d_{50} is the mean particle size.

Fig. 19 shows the correlation between the stress normalized shear modulus $G_0/(p_0)^{1/3}$ and modified coordination number CN' for five different particle shapes. The coefficient of determination R^2 is $0.[(F_i/F_{\text{avg}})(R_i^*/d_{50})]^{1/3} 89$, which is better than those in Fig. 13 and Fig. 14 $[(F_i/F_{\text{avg}})(R_i^*/d_{50})]^{1/3}$. The result suggests that the coordination number and the distribution of normal contact stiffness could be the dominant factors for the elastic stiffness of granular materials regardless of particle shape. The proposed modified coordination number can better relate the microscopic properties with the macroscopic properties.

Fig. 20(a) compares the secant modulus ratio G/G_0 and contact sliding ratio f_s for specimens with different particle shapes. The contact sliding ratio increases dramatically when the strain level exceeds 1.0×10^{-5} while the secant modulus ratio starts to decrease after the shear strain reaching $\gamma = 1.0 \times 10^{-4}$. An inverse relationship between the modulus ratio and the contact sliding ratio can be observed.

Modulus ratio depends on the collective inter-particle behavior. To account for the effect of contact stiffness distribution, the same weighted factor $[(F_i/F_{\text{avg}})(R_i^*/d_{50})]^{1/3}$ is adopted for each contact to account for its significance, and the relationship between the secant modulus ratio and modified sliding ratio is presented in Fig. 20(b). The result shows that the modified contact sliding ratio remains zero when the strain level is less than 1.0×10^{-5} and it starts to increase after the strain level exceeds 1.0×10^{-5} , but the secant shear modulus starts to decrease after a higher shear strain (i.e., $\gamma = 1.0 \times 10^{-4}$) was reached. Moreover, G/G_0 keeps decreasing after the shear strain γ reaching 5.0×10^{-4} when the contact sliding ratio remains almost stable, because the modified contact sliding ratio is directly related to the tangential shear modulus rather than the secant shear modulus. Therefore, the evolution tangential modulus ratio G_{tan}/G_0 with shear strain is plotted in Fig. 20(c) in conjunction with the modified contact sliding ratio f_s' . The result indicates that the tangential modulus ratio G_{tan}/G_0 starts to decrease at γ of 1.0×10^{-5} and then remains almost constant after shear levels exceeds 5.0×10^{-4} which shows a better correspondence.

4. Summary and Conclusions

In this study, five different particle shapes were considered to investigate the effect of macroscopic and mesoscopic particle shape on small strain stiffness G_0 , modulus ratio G/G_0 , and damping ratio D using DEM simulations. The microscopic properties for assemblies with different particle shapes and shear strains were explored simultaneously and a weighted factor is proposed to account for the effect of contact stiffness distribution. The major findings are summarized as follows:

- 1) The modulus ratio G/G_0 increases and the damping ratio D decreases with increasing confining pressure p_0 . However, damping ratios at a relatively low confining pressure can be smaller than those at a higher confining pressure due to the dilatancy of the specimen.
- 2) Shear modulus G increases with decreasing void ratio, but the void ratio has no significant effect on G/G_0 . Additionally, no significant effect of void ratio can be found on D at a relatively small strain level. However, the void ratio affects the damping ratio at a larger strain level (e.g., 1×10^{-3}) due to the contraction or dilatancy of the specimen.

- 3) Low regularity results in a higher shear modulus G due to the decrease of rattles, while macroscopic and mesoscopic morphologies of particles have no significant effect on the stress exponents n and modulus ratio G/G_0 . Nevertheless, particle shape affects the dilatancy of specimens. This leads to the different values of damping ratios among different particle shapes at relatively larger strain level.
- 4) The relationship between stress normalized shear modulus and coordination number as well as tangential modulus ratio and contact sliding ratio can be improved by considering the effect of contact stiffness distribution.

CRediT authorship contribution statement

Jing Hu: Conceptualization, Methodology, Validation, Writing – review & editing, Funding acquisition. **Hongwei Wu:** Conceptualization, Investigation, Software, Writing – original draft. **Xiaoqiang Gu:** Conceptualization, Methodology, Supervision, Writing – review & editing, Funding acquisition. **Qihui Zhou:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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